

CSF2702 – Lab Book

This Lab Book contains the lab tasks for CSF2702 which are designed to be completed during the lab class each week.

Each week you will be given a new lab class sheet with new tasks. You should aim to get all tasks completed and marked off by staff in the lab class. If you do not get the tasks finished in the lab class then you should have them completed for the next lab session.

Full marks will be awarded if the tasks are marked off by staff during the session they are handed out; or if they are marked off by staff during the following session. Half marks will be awarded if the tasks are signed off later than this.

The lab work as a whole is worth 15% of the module (1.5% per lab).

CSF2702 – Lab Class 1 –

☐ Task 1.1

This task involves writing two Java classes. The purpose of this task is to recap and get you familiar with Java and objects once again.

We want to model two entities: A person and a lecturer. There is an inheritance relationship between the two: decide which is the superclass and which is the subclass.

A Person has the following attributes:

- Name (String), this is mandatory.

A lecturer has the following **additional** attributes:

- Staff ID (Integer).
- Office location (String), For example “Computational Foundry 413” .

Create Java classes to model these two entities. You should only be able to create a person if you provide their name. Similarly, you should only be able to create a lecturer if you provide their name, staff ID and their office location.

Your classes should have setters and getters for all fields (unless you can think of specific reasons for not allowing them to be set after creation).

Place a print statement in each constructor so that you can see the constructors being executed and in which order they are executed in order.

Test your classes in your main method. You should create at least two objects: a person and a lecturer and play around with the setters and getters (and print statements) to make sure they work as expected.

□ Task 1.2

The office location in Task 1.1 is a bit simplistic. Really speaking this should be an additional class with at least the following two instance variables:

- Building (e.g., “Margam Building”)
- Room (E.g., “CoSMOS”)

Create such a class called “OfficeLocation”, and then alter the lecturer class so that it uses your new class. Remember that instances of this class can be passed as parameters in constructors and methods.

□ Task 1.3

In the Person class, create a ‘printDetails’ method which prints all the necessary information (in this case, only the person’s name). Override this ‘printDetails’ method in the Lecturer class to display all the necessary information.

What happens when you call this method through a superclass reference? For example, if you chose Person as your superclass and Lecturer as your subclass:

- Person p = new Lecturer();
- p.printDetails();

Does this behave the way you expected?

What happens when you call a method from the subclass:

- Person p = new Lecturer();
- p.getStaffID();

What happens and why?